

```
(venv) nipurna@AK-CPU:~/Desktop/GenAI/Practicals/milestone-02-rag-application$ python
-m pytest -v -s tests/test_evaluation_15.py
```

=== test session starts

=====

platform linux -- Python 3.10.12, pytest-9.0.3, pluggy-1.6.0 --

```
/home/nipurna/Desktop/GenAI/Practicals/milestone-02-rag-application/venv/bin/python
```

cachedir: .pytest_cache

```
rootdir: /home/nipurna/Desktop/GenAI/Practicals/milestone-02-rag-application
```

plugins: anyio-4.13.0, langsmith-0.8.5

collecting ... Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

Loading weights:

100%

103/103 [00:00<00:00, 6699.13it/s]

collected 1 item

tests/test_evaluation_15.py::test rag fifteen questions evaluation

[EVALUATION TEST 1/15]

QUESTION : What was the main problem with sequential processing before transformers?
NOVA AMZN : The main problem with sequential processing before transformers was that it worked like a person reading a book through a straw, where they could only see one word at a time. By the time they reached page 50, they had a very blurry memory of what happened on page 1. This method was slow and forgetful, as it didn't see the entire context of the sentence or document at once, leading to a loss of earlier information as it progressed.

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[EVALUATION TEST 2/15]

QUESTION : Explain the 'Satellite View' analogy used to describe how a transformer works.

NOVA AMZN : The 'Satellite View' analogy is used to describe how a transformer processes information differently from traditional methods. Here's a detailed explanation:

The Problem with the Old Way: The Single-File Line

Before transformers, AI systems processed information sequentially, much like a person reading a book through a straw. They could only focus on one word at a time, similar to how a straw lets you drink one sip at a time. By the time they reached the middle of the book, they had a very blurry memory of what happened at the beginning. This method of sequential processing was slow and prone to forgetting earlier context because each new piece of information was processed in isolation.

The Transformer Solution: The Satellite View

A transformer works like a satellite taking a high-resolution photo of an entire city at once.

Unlike the single-file line approach, where context is lost over time, a transformer views the

entire sentence or document all at once. This means it doesn't read left to right word by word but processes every word in the context of the entire document simultaneously. This allows it to understand the relationships, meaning, tone, and facts of each word in the context of the whole, much like how a satellite sees the whole city and understands spatial relationships and patterns.

In essence, the 'Satellite View' analogy highlights the transformer's ability to process and understand context in a holistic manner, rather than sequentially. This allows it to make more informed and accurate predictions and generate responses that maintain coherence and context throughout the entire input.

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[EVALUATION TEST 3/15]

QUESTION : How does self-attention handle the relationship between words like 'it' and 'phone'?

NOVA AMZN : Self-attention in the model works by "highlighting" or emphasizing the most relevant connections between words in a sentence. In the example provided, "The battery of the phone died because it was old," self-attention allows the model to draw a strong connection between the word "it" and the word "phone."

In this context, "it" refers to the "phone," not the "battery," because a phone can become old over time, whereas a battery does not typically get "old" in the same way. The model uses attention mechanisms to mathematically determine which words are most relevant to each other, ignoring less relevant words like "battery" in this particular relationship.

This process allows the model to understand complex relationships and dependencies in a sentence, recognizing that certain words are more closely related than others based on the context. This is akin to how a human reader would naturally focus on the most pertinent parts of a sentence when trying to understand its meaning.

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[EVALUATION TEST 4/15]

QUESTION : What are the specific roles of Expert 1, Expert 2, and Expert 3 in Multi-Head Attention?

NOVA AMZN : In the Multi-Head Attention framework, each expert has a specific role to play in understanding the text:

- **Expert 1**: This expert focuses solely on grammar. They analyze the sentence structure, punctuation, and overall grammatical correctness to ensure that the text adheres to standard linguistic rules.

- **Expert 2**: This expert is dedicated to understanding the emotions conveyed in the text. They pay attention to words and phrases that express feelings, moods, and emotional undertones, helping to capture the emotional context of the text.

- **Expert 3**: This expert looks specifically for dates and numbers within the text. They identify and interpret any temporal or quantitative information, which is crucial for understanding events, timelines, and factual data.

By working together, these experts provide a comprehensive understanding of the text by covering different aspects such as grammar, emotions, and factual data.

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[EVALUATION TEST 5/15]

QUESTION : Why do transformers need Positional Encoding, and what analogy is used for it?

NOVA AMZN : Transformers need Positional Encoding because, unlike traditional models that process sentences sequentially (one word at a time), they process the entire sentence at once. This means they can't inherently understand the order of the words since they see all the words simultaneously. To fix this, transformers use Positional Encoding, which is akin to giving every word a "tag" that includes its position in the sentence. This allows the model to understand the sequence and order of words, much like how a person reading a book through a straw (sequentially) understands the order of words by processing one at a time.

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[EVALUATION TEST 6/15]

QUESTION : What is the difference between the Encoder (The Chef) and the Decoder (The Waiter)?

NOVA AMZN : The difference between the Encoder (The Chef) and the Decoder (The Waiter) in the context of a transformer model can be understood through their respective roles in the process:

Encoder (The Chef):

- The Encoder takes the raw input (your input words).
- It processes these words to understand their meaning and context.
- It transforms the input into a concentrated "sauce" or a numerical map that represents the overall meaning of the input sequence.
- Think of it as chopping up the ingredients and creating a base that holds the essence of the original sentence.

Decoder (The Waiter):

- The Decoder takes the concentrated "sauce" produced by the Encoder.
- It uses this numerical map to generate a coherent response or output.
- It looks at what it has already generated to decide what to serve next, ensuring that the output makes sense as a whole.
- Imagine it as the person who takes the concentrated base and turns it into a finished dish, considering what has already been served to create a complete meal.

In essence, the Encoder understands the input, while the Decoder uses this understanding to produce the output.

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[EVALUATION TEST 7/15]

QUESTION : How does a transformer model learn during the training phase?

NOVA AMZN : A transformer model learns during the training phase by playing a game of "Fill in the Blanks." Researchers provide the model with billions of sentences where certain words are hidden behind a gray box. The model then guesses the missing word and receives feedback on whether its guess is correct. Based on this feedback, the model adjusts its internal settings. This process is repeated trillions of times, allowing the model to become highly adept at predicting the next word in a sentence. Through this extensive practice, the model develops an understanding of complex sentence structures, meanings, tones, and contextual facts, which enables it to appear as though it "knows" a wide range of information from various domains.

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[EVALUATION TEST 8/15]

QUESTION : What is LangChain and what is the 'LEGO connectors' analogy describing?

NOVA AMZN : LangChain is a toolkit designed to simplify the process of building software powered by large language models. While these models can generate text independently, they often lack the ability to access real-time data, remember long conversations, or interact with external software. LangChain provides the necessary connections to bridge these gaps by linking models to other data sources and tools.

The 'LEGO connectors' analogy describes how LangChain functions as the central framework that allows developers to connect various components together to create a cohesive application. Just like LEGO pieces can be linked in different ways to build various structures, LangChain's Chains module lets developers link multiple components—such as input processing, model interactions, and output formatting—to create a single, functional application. This modular approach enables flexibility and customization in building sophisticated software applications.

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[EVALUATION TEST 9/15]

QUESTION : List the four sequential steps a typical LangChain 'chain' might perform.

NOVA AMZN : A typical LangChain 'chain' might perform the following four sequential steps:

1. **Input Handling**: Takes user input and processes it.
2. **Prompt Formatting**: Formats the processed input into a specific prompt suitable for the language model.
3. **Model Interaction**: Sends the formatted prompt to the language model for generating a response.
4. **Output Parsing**: Parses the model's output into a specific format like a list or a JSON object for final delivery to the user.

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[EVALUATION TEST 10/15]

QUESTION : What are the three major weaknesses of LLMs that LangChain fixes?

NOVA AMZN : LangChain addresses three major weaknesses of large language models (LLMs):

1. ****Lack of Access to Real-Time Data****: LLMs are typically trained on static datasets and cannot access real-time data. LangChain provides the necessary connections to external data sources, enabling the model to use up-to-date information.
2. ****Inability to Remember Long Conversations****: Most language models are stateless, meaning they treat every query as a brand-new interaction and forget previous messages. LangChain's Memory module stores past interactions, allowing the model to refer back to earlier parts of a conversation.
3. ****Limited Access to External Data****: LLMs cannot access specific data that was not part of their original training, such as private documents or specific texts. LangChain's Retrieval module facilitates Retrieval Augmented Generation, enabling the model to access and use relevant external data in responses.

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[EVALUATION TEST 11/15]

QUESTION : Identify two common locations or enterprise use cases where LangChain is used.

NOVA AMZN : LangChain is often used in two common locations or enterprise use cases:

1. ****Customer Support and Chatbots****: LangChain can be integrated into customer support systems to create more intelligent and efficient chatbots. By utilizing the Retrieval module, the chatbot can access specific data from company documents, FAQs, or customer databases to provide accurate and relevant responses. The Memory module allows the chatbot to remember previous interactions with the same customer, leading to a more personalized and coherent conversation.
2. ****Content Generation and Research****: In research or content creation, LangChain can be used to augment large language models with external data sources. For example, a research assistant powered by LangChain can pull in specific, up-to-date information from scientific papers, company reports, or proprietary databases to generate detailed reports, summaries, or analyses. The Retrieval module helps in fetching the most relevant and recent documents, while the Memory module can keep track of the context of the ongoing research to ensure continuity and coherence in the generated content.

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[EVALUATION TEST 12/15]

QUESTION : When should a developer move past a prototype and start using LangChain?

NOVA AMZN : A developer should consider moving past a prototype and start using LangChain when they need to build a more complex, scalable, and interactive application

that leverages large language models. Specifically, the following scenarios indicate that it's time to transition:

1. ****Need for Real-Time Data Integration****: If the prototype requires accessing real-time data or integrating with external APIs and databases, LangChain's Chains module can facilitate these connections.
2. ****Conversational Continuity****: If the prototype lacks the ability to maintain context over multiple interactions, the Memory module in LangChain can be used to store past interactions and enable more coherent and contextually aware conversations.
3. ****Access to External Knowledge****: If the prototype needs to access specific external data sources that were not part of the language model's training, the Retrieval module can be employed to fetch and incorporate relevant information dynamically.
4. ****Complex Workflows****: If the prototype involves a series of interconnected tasks or steps that need to be executed in a specific order, LangChain's Chains can help in orchestrating these steps into a coherent workflow.
5. ****Dynamic Agent Capabilities****: If the prototype requires more intelligent and adaptive behavior from the model, LangChain's Agents module can provide the necessary flexibility and dynamism.

In essence, when the prototype starts to show limitations in handling complex interactions, real-time data, or when it needs to maintain context and access external information, it's time to leverage LangChain for a more robust and sophisticated application.

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[EVALUATION TEST 13/15]

QUESTION : Explain the 'Open Book exam' analogy regarding LangChain's Retrieval (RAG) process.

NOVA AMZN : The 'Open Book exam' analogy is a useful way to understand LangChain's Retrieval Augmented Generation (RAG) process. In traditional closed-book exams, students must rely solely on their memory to answer questions. However, in an open-book exam, students can refer to any materials they have prepared, allowing them to provide more accurate and detailed answers.

In LangChain's Retrieval (RAG) process, the model functions similarly to a student in an open-book exam. When a user asks a question, the Retrieval module acts like a student consulting reference materials. It doesn't just rely on the pre-trained knowledge within the model but also accesses additional data sources—such as private documents, databases, or external texts—that were not part of its original training.

Here's how it works in detail:

1. ****Question Analysis****: The model first analyzes the user's question to understand what information is needed.

2. **Data Retrieval**: The Retrieval module searches through its stored database of documents, breaking them into small, relevant pieces and identifying the most pertinent information.
3. **Context Integration**: This retrieved information is then integrated into the model's response, much like how a student might highlight key points in their reference material to answer the question accurately.
4. **Response Generation**: Finally, the Decoder (or 'Waiter') uses this augmented information to generate a comprehensive and precise response.

In essence, the 'Open Book exam' analogy highlights how LangChain's Retrieval module enhances the model's ability to provide accurate and detailed answers by allowing it to consult additional, external data sources during the query process.

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[EVALUATION TEST 14/15]

QUESTION : What is an Agent in LangChain and how does the 'smart intern' analogy apply to it?

NOVA AMZN : An Agent in LangChain is a more dynamic and flexible component compared to standard chains. Instead of following a strict sequence of steps, an agent leverages the language model as a reasoning engine to determine the most appropriate actions to take in response to a query. For example, if an agent is given access to a search engine and a calculator, it can autonomously decide whether it needs to look up a fact or perform a mathematical operation to provide an answer.

The 'smart intern' analogy applies to an Agent in LangChain in the following way: Just like a smart intern who can choose the best tools and resources to complete a task efficiently, an Agent uses its access to various tools and data sources to determine the best course of action dynamically. It doesn't follow a fixed set of instructions but instead assesses the situation and uses its capabilities—such as searching for information or performing calculations—as needed to achieve the best outcome. This flexibility allows Agents to handle a wide range of queries more effectively than a rigid, step-by-step approach.

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[EVALUATION TEST 15/15]

QUESTION : How does LangChain serve as a 'universal remote' for different AI models?

NOVA AMZN : LangChain serves as a "universal remote" for different AI models by providing the necessary infrastructure to connect these models to various data sources and tools, thus enabling them to perform more complex tasks that go beyond their original capabilities.

Here's how it achieves this:

1. **Chains**: LangChain's Chains module allows different components to be strung together in a sequence to perform a task. This flexibility means you can create custom workflows that can integrate various AI models and external tools.
2. **Memory**: The Memory module in LangChain helps maintain context over multiple interactions. Since most language models are stateless, this module allows them to

remember past interactions, making them capable of maintaining context in conversations and improving the coherence and continuity of responses.

3. **Retrieval**: The Retrieval module is designed for Retrieval Augmented Generation (RAG). It enables the model to access external data sources that were not part of its original training. This allows the model to fetch relevant information from databases, documents, or other data stores, thereby enhancing its ability to provide accurate and up-to-date information.

4. **Agents**: LangChain's Agents module allows for more dynamic and adaptive interactions. Agents can use the language model as a reasoning engine to determine which actions to take based on the current context. This means they can decide whether to look up information or perform calculations on their own, thus acting more like a "universal remote" that can handle a wide range of tasks without predefined steps.

By integrating these modules, LangChain provides a comprehensive framework that can seamlessly connect and leverage different AI models, making it a powerful tool for building applications that require complex interactions with external data and tools.

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